Dobrushin's Wilson criterion: an explicit strong-coupling gap at $g^2 > 444$ and an open block test

The finite-volume criteria of the finite-certificate treatment have a single-site ancestor, Dobrushin's uniqueness condition (Theory Probab. Appl. 13 (1968) 197), and for the Wilson action it can be checked by hand. The comparison below uses four-dimensional lattices and spatial tori with sides at least four, as in the polymer note. The conditional distribution of one link given all the others is

$$d\mu_\ell(U \,|\, \omega) \;\propto\; \exp\Big[\frac{\beta_W}{N}\,\mathrm{Re\,tr}\,(U\,M_\ell(\omega))\Big] d\mu(U), \qquad M_\ell = \sum_{p \ni \ell} S_p,$$

the sum of the six staples of ℓ in four dimensions. Changing one neighbouring link changes one staple, and the resulting total-variation distance between the two conditional distributions is at most $e^{4\beta_W} - 1$, independently of N. Each link has 18 neighbours that share a plaquette with it, so Dobrushin's coefficient is

$$\alpha = \sup_\ell \sum_{\ell'} \rho_{\ell\ell'} \;\le\; 18\big(e^{4\beta_W} - 1\big),$$

and the condition $\alpha < 1$ holds for $\beta_W < \frac{1}{4}\log\frac{19}{18} \simeq 0.01352$, that is

$$g^2 > \frac{8N}{\log(19/18)}; \qquad g^2 > 148N = 444 \quad (SU(3)) \text{ is a sufficient rounded bound.}$$

Dobrushin's theorem gives a unique Gibbs state, Künsch's theorem (Commun. Math. Phys. 84 (1982) 207) gives exponential decay of all truncated correlations at rate $\log(1/\alpha)$ per lattice unit, and link reflection positivity turns the rate into a gap:

$$\Delta_W \;\ge\; \frac{\hbar c}{a}\log\frac{1}{18(e^{24/g^2} - 1)} \;\simeq\; \frac{\hbar c}{a}\log\frac{g^2}{432} \qquad (g^2 > 444),$$

uniformly in the volume. This is a better threshold than the polymer expansion's sufficient integer bound 1059 (1056 is linearized). Its rate grows like $\log(g^2/432)$, whereas the polymer rate grows like $4\log(g^2/1056)$ at large g^2. The block version of the same condition, the Dobrushin–Shlosman constructive criterion (in *Statistical Physics and Dynamical Systems*, Birkhäuser 1985, 347), replaces the single link by a box V of links and is expected to hold deeper into the intermediate region as V grows; it is the quantity a finite verification would compute, and Section 4 states it as a specification. All references at metadata level; constants explicit; nothing promoted.

Correction, 2026-10-02 (comparison rates). The earlier table printed the leading logarithms as rigorous lower bounds. The exact exponential-activity rates are given in the corrected table in §3; the logarithms are asymptotic estimates. This also distinguishes the exact single-link condition from its rounded sufficient threshold.

1. The one-link conditional distribution

Fix all links except ℓ. The plaquettes containing ℓ are six in four dimensions, and each contributes $\mathrm{Re\,tr}(U_\ell S_p)$ with $S_p \in SU(N)$ the product of the other three links of p in the appropriate order. Hence the conditional density is $e^{f_\omega(U)}$ with $f_\omega(U) = (\beta_W/N)\,\mathrm{Re\,tr}(UM_\ell)$, $M_\ell = \sum_{p\ni\ell} S_p$, normalized by $Z(\omega) = \int e^{f_\omega} d\mu$. Gauge invariance of the interaction is visible in the fact that a gauge transformation at an endpoint of ℓ rotates M_ℓ and the Haar measure absorbs it; nothing more is needed.

2. The Dobrushin coefficient

Lemma 1. For probability densities $p \propto e^{f}$, $q \propto e^{f'}$ with respect to the same measure, $\|p-q\|_{\mathrm{TV}} \le e^{\operatorname{osc}(f-f')}-1$.

Proof. $p/q = e^{f-f'}Z_{f'}/Z_f \le e^{\sup(f-f')}e^{\sup(f'-f)} = e^{\operatorname{osc}(f-f')}$, so $p-q \le (e^{\operatorname{osc}}-1)q$ and $\|p-q\|_{\mathrm{TV}} = \int (p-q)_+ \le e^{\operatorname{osc}}-1$. □

Lemma 2. If ω, ω' differ only at a link ℓ' sharing a plaquette with ℓ, then $\operatorname{osc}_U(f_\omega - f_{\omega'}) \le 4\beta_W$.

Proof. One staple changes, $M_\ell - M'_\ell = S_p - S'_p$ with $S_p, S'_p \in SU(N)$, so $|f_\omega(U) - f_{\omega'}(U)| = (\beta_W/N)|\operatorname{Re}\operatorname{tr}(U(S_p - S'_p))| \le (\beta_W/N)\|S_p - S'_p\|_1 \le (\beta_W/N)\cdot 2N$, the trace norm of a difference of two unitaries being at most $2N$; the oscillation is at most twice the supremum. □

If ℓ' shares no plaquette with ℓ, the conditional distribution at ℓ does not depend on it. Each link lies in six plaquettes with three further links each, and the six plaquettes share no link besides ℓ, so there are exactly 18 links ℓ' with $\rho_{\ell\ell'} \neq 0$. Hence

$$\alpha = \sup_{\ell} \sum_{\ell'} \rho_{\ell\ell'} \le 18\big(e^{4\beta_W} - 1\big),$$

and the upper bound on α is strictly below one precisely when $\beta_W < \tfrac{1}{4}\log(19/18)$. The reported 0.01352 and $148N$ are rounded figures; $g^2 > 148N$ is sufficient since $\log(19/18) > 2/37$.

3. From the condition to the gap

Dobrushin's theorem: $\alpha < 1$ implies a unique Gibbs state on $\mathbb{Z}^4$ and on every torus. The covariance estimate of Föllmer (J. Funct. Anal. 46 (1982) 387; Künsch, loc. cit., for the decay statements): under $\alpha < 1$, for local observables F, G,

$$|\langle F; G\rangle| \;\le \sum_{\ell \in \operatorname{supp} F,\ \ell' \in \operatorname{supp} G} \delta_\ell F \,\big[(1-\rho)^{-1}\big]_{\ell\ell'}\, \delta_{\ell'} G,$$

with $\delta_\ell F$ the oscillation of F in the link ℓ, and since $\rho^n_{\ell\ell'}$ vanishes unless the link distance is at most n lattice units, $[(1-\rho)^{-1}]_{\ell\ell'} \le \alpha^{d(\ell,\ell')}/(1-\alpha)$. Truncated correlations at time separation T therefore decay at least like $\alpha^{T/a}$, rate $m = \log(1/\alpha)/a$. The Wilson transfer matrix is positive by link reflection positivity, and for a local A orthogonal to the vacuum $\langle A, \mathcal{T}^n A\rangle \le C_A \alpha^n$ forces its spectral measure to vanish above α, so on the cyclic subspace of local observables

$$\Delta_W = -\frac{\hbar c}{a}\log\|\mathcal{T}|_{\Omega^\perp}\| \;\ge\; \frac{\hbar c}{a}\log\frac{1}{\alpha} \;\ge\; \frac{\hbar c}{a}\log\frac{1}{18(e^{24/g^2}-1)} \qquad (SU(3)).$$

route	rigorous threshold ($SU(3)$)	gap bound
polymer expansion, all representations	$g^2 \ge 1059$ (sufficient integer bound)	$(\hbar c/a)\,4\log[1/(176(e^{6/g^2}-1))]$
Dobrushin single-site condition	$g^2 > 24/\log(19/18)$; $g^2 > 444$ suffices	$(\hbar c/a)\,\log[1/(18(e^{24/g^2}-1))]$

The two are complementary: the Dobrushin route enters earlier, the polymer route's rate is larger at sufficiently large coupling, rather than throughout their overlap. For $U(1)$ the same computation gives $g^2 > 148$ with $N = 1$.

4. The block criterion, as a specification

A tempting block analogue uses the single-link marginal influence

$$\rho_V(x,y) = \sup_{\omega,\omega' \text{ differ only at } y} \|\mu_V(\cdot \mid \omega)|_x - \mu_V(\cdot \mid \omega')|_x\|_{\mathrm{TV}}$$

and tests $\sum_{x\in V, y\notin V} \rho_V(x,y) < |V|$. For a link with an interior endpoint, gauge invariance makes its marginal Haar for every exterior boundary, so this influence is zero. A condition that only sees these marginals cannot certify the joint mixing needed by the Wilson-transfer argument. This does not empty the single-site criterion proved above: there all other links are conditioned on and that interior gauge transformation is unavailable.

A usable block certificate must control the whole relevant invariant joint marginal, including boundary-attached paths, not just selected small Wilson loops. Applicability of a finite-size theorem, quantitative slack, robustness in the generated-interaction norm and all-volume uniformity are stated in the canonical certificate treatment. The often suggested sides 3–5 or 10–30 are trial estimates, not certified necessary or sufficient sizes. Dobrushin–Shlosman's finite-size results are source leads at metadata level here, not a checked theorem for this proposed gauge-block test.

5. Consequence for STATE

The proved one-link criterion gives the displayed sufficient fixed-cutoff Wilson region and rate; its comparison with the polymer bound uses the exact rates in §3. A joint block certificate and weak-side multiscale control remain open. No smallest certified crossover coupling, monotone coupling interval or reduction to two scalar numbers has been established. Kogut–Susskind stability and the physical continuum gap are separate.